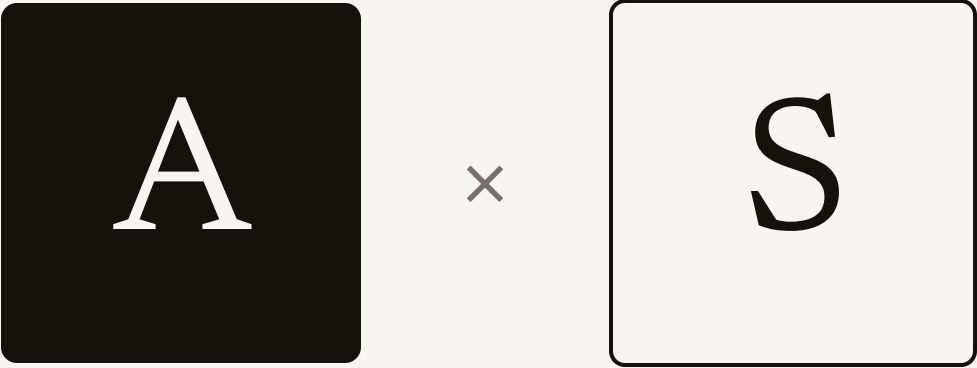




Apple vs. Samsung

A Strategic Economic Analysis



COURT

APPLE

SAMSUNG

SAMSUNG

Apple won in court. *Samsung won the market.*

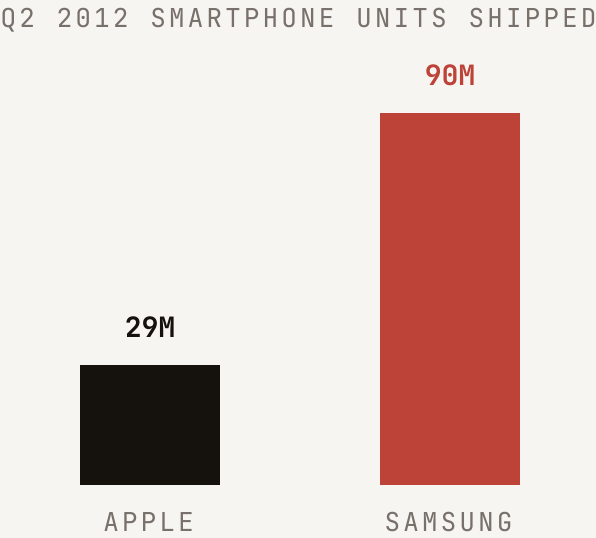
How can both be true?



AUGUST 2012 – VERDICT

\$1.05B

LARGEST PATENT VERDICT IN U.S. HISTORY



FROM QUESTION TO VERDICT

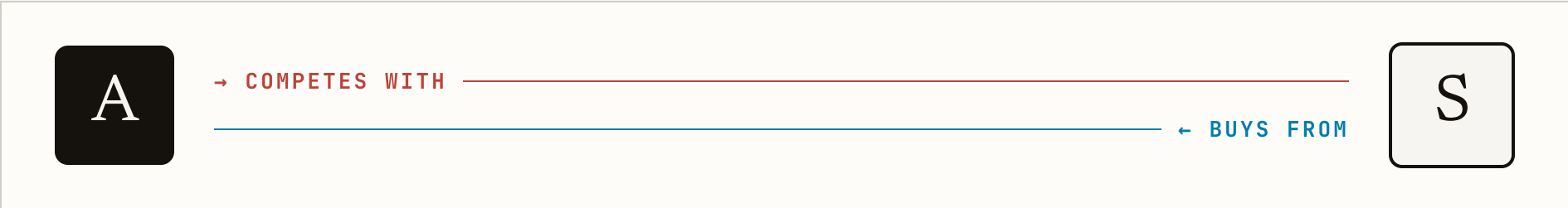
Our argument in *six beats*.

I	The Setup Why they couldn't just walk away.	SLIDE 5
II	Three Pressures Why Apple actually sued.	SLIDES 7-10
III	The Pivot Why the threat had a ceiling.	SLIDE 11
IV	The Logic Two parallel games at once.	SLIDES 12-13
V	The Evidence What investors actually saw.	SLIDES 14-15
VI	The Verdict Who won — and what it means.	SLIDE 16

I · ONE STRANGE RELATIONSHIP

Rivals *and* supplier–customer at the same time.

This duality — **coopetition** — constrains every move that follows in the dispute.



FACT	NUMBER
Samsung's share of iPhone manufacturing cost	26%
Apple's component purchases from Samsung (2010)	\$6B
Apple's share of Samsung's total revenue	4%

"Apple was reluctant to take legal action against Samsung, given its key role in supplying chips and components."

— WESLEY CASE STUDY

II • APPLE DID NOT SUE OUT OF ANGER

Three economic forces pushed Apple *into court*.

Each one is the subject of one of the next three slides.

1 PRESSURE #1

Patent Hold-Up

Too many patents, no clean way out — and the standard fix (cross-licensing) had failed.

2 PRESSURE #2

Lost Differentiation

Samsung looked too much like Apple, and Apple's premium was eroding.

3 PRESSURE #3

Platform Tipping

Android was running away with the smartphone OS market — second place might not survive.

IF YOU REMEMBER NOTHING ELSE FROM THIS DECK — REMEMBER THESE THREE.

WHY APPLE SUED – I

When no one owns the road, everyone pays a toll.

~10K patents needed
TO SHIP ONE SMARTPHONE

\$10B / year, industry-wide
ON PATENT LITIGATION BY 2011

Apple & Google each spent *more on lawsuits than R&D*.
– ECEB SECTION 3.1

Cross-licensing is the textbook fix. It failed in Aug 2010.

PATENT STRATEGY – COMPARED

	APPLE	SAMSUNG
Offensive	Sue Samsung for copying iPhone	Counter-sue over wireless patents
Defensive	"Patent everything" — wall of patents	Build own patent stockpile
Bargaining chip	Win cases → better licensing terms	Wireless patents as leverage

WHY APPLE SUED - II

Bertrand Competition & Hotelling, in plain sight.

BERTRAND COMPETITION

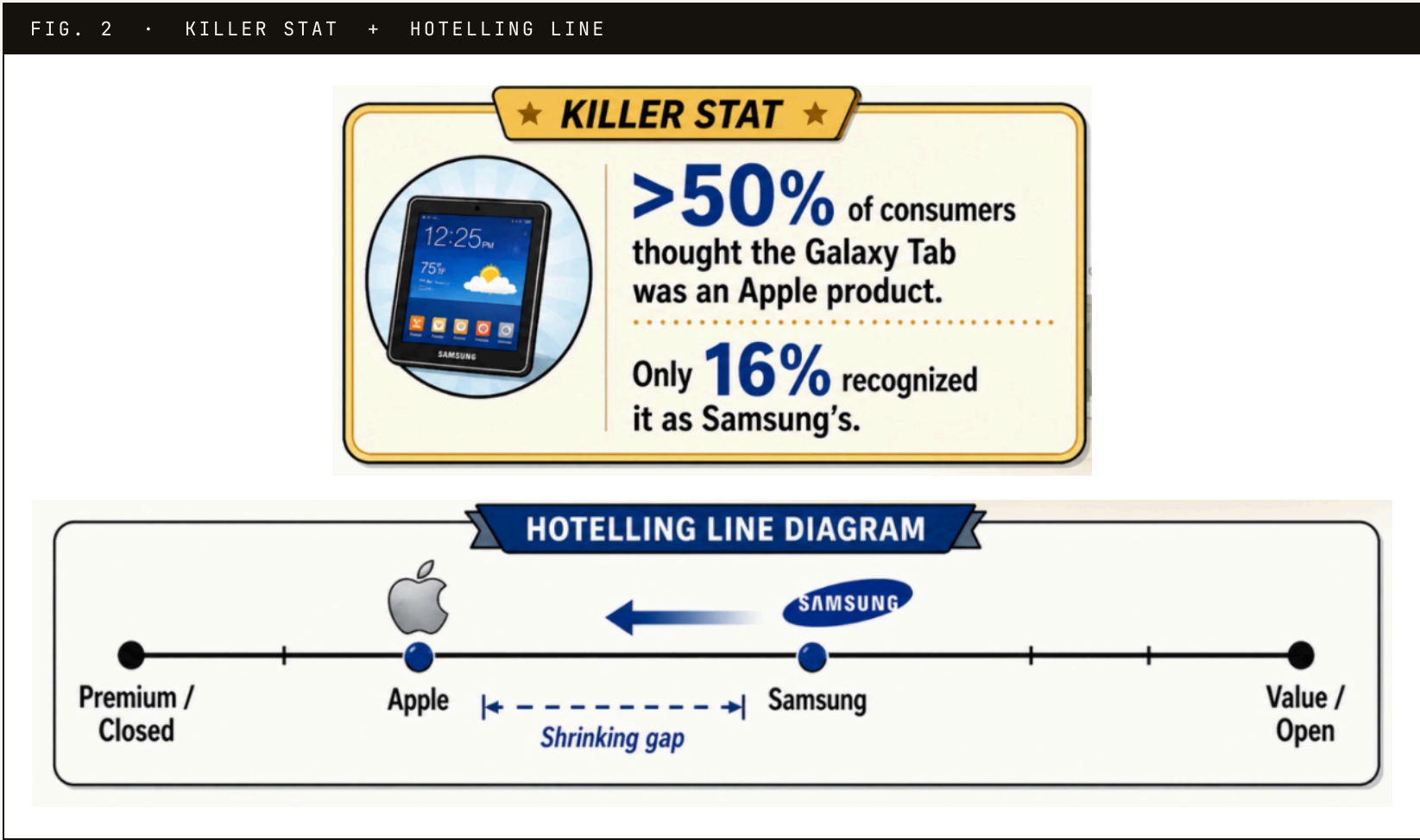
Identical products → price war → zero profit.

HOTELLING

The closer rivals stand on the "taste line," the harder they compete.

IMPLICATION

Samsung's copying moved it toward Apple → eroded the premium.



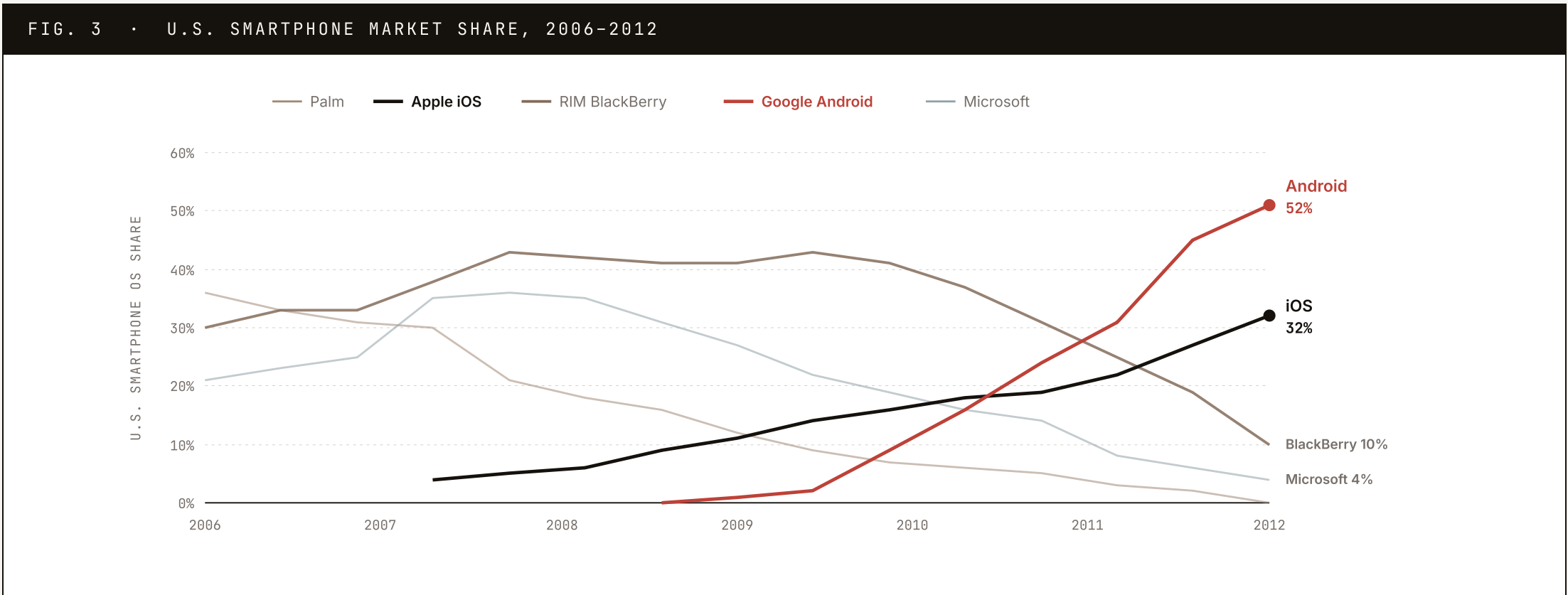
CAPTION: AS SAMSUNG'S POSITIONING APPROACHED APPLE'S, THE DIFFERENTIATION PREMIUM ERODED.

WHY APPLE SUED – III

Network effects don't reward second place.

TWO PLATFORMS, TWO GAMES

	IOS (APPLE)	ANDROID (SAMSUNG+)
Standard	Closed	Open
Network effect	Apps → users	Same — across many OEMs
Switching costs	High (apps, iCloud)	Low between brands
Game	Competes FOR the market	Competes IN the market



SOURCE: CASE EXHIBIT 10 – CANALYS, SILICON VALLEY INSIDER, COMSCORE, GARTNER.

III · THE PIVOT – APPLE'S THREAT HAD A CEILING

"I will spend my last dying breath... every penny of Apple's \$40 billion in the bank... I'm going to destroy Android."

– STEVE JOBS

STRATEGIC COMMITMENT – 3 CRITERIA

Observable?

Public lawsuit + injunctions sought

✓ YES

Understandable?

Copy us → face \$1.05B+ costs

✓ YES

Credible?

Sunk costs, reputation on the line

△ PARTIAL

WHY ONLY PARTIAL?

PUSH TO SUE

Already suing **HTC, Motorola, Nokia, Amazon** and **Google** (Exhibit 2) → softening on Samsung breaks every other threat.

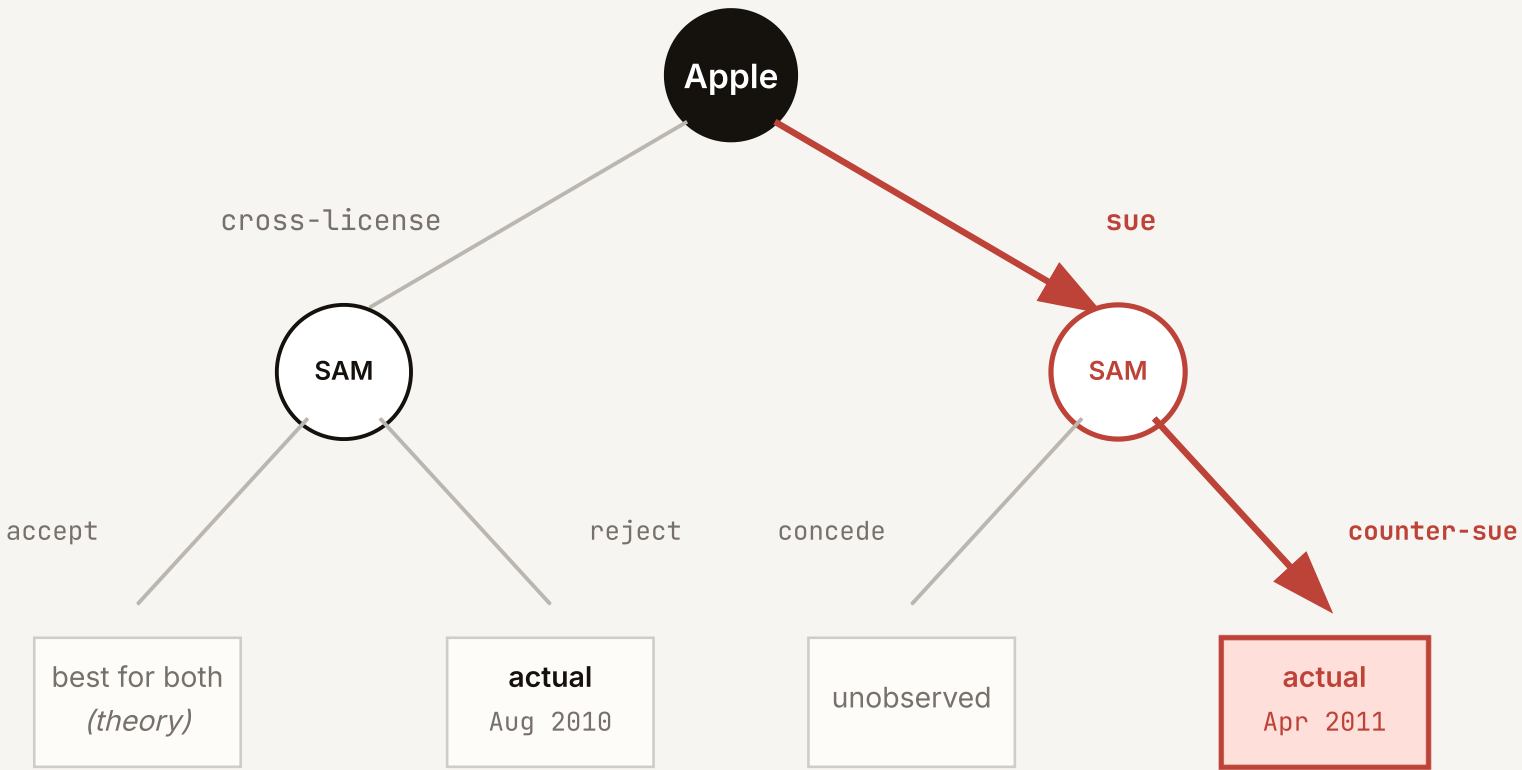
trigger strategy / reputation in a repeated game · suit is the minimum-viable move

PULL TO HOLD BACK

Samsung supplies Apple's chips, screens, memory → can't actually push to ban.

coopetition / hold-up

GAME TREE – HIGHLIGHTED: ACTUAL PATH



CROSS-LICENSE → REJECT → SUE → COUNTER-SUE

APPLE'S THREAT LACKED FULL CREDIBILITY – BOTH SIDES SETTLED IN COURT.

IV · COOPETITION LOGIC

Apple and Samsung play *two games at once*
— and the second one saved the relationship.

GAME 1 Smartphone Competition

PRISONER'S DILEMMA

	SAM: DIFFERENTIATE	SAM: IMITATE
APPLE: TOLERATE	Mod π , Mod π	Low π , High π
APPLE: SUE	–Legal, 0	–Legal, –Legal

GAME 2 Supply Chain Cooperation

REPEATED GAME

	SAM: RELIABLE SUPPLY	SAM: DISRUPT
APPLE: CONTINUE	★ \$6B+, \$6B+ NE	Crippled, Short gain
APPLE: DIVERSIFY	Transition cost, Lose customer	Both lose, Both lose

COOPERATION INEQUALITY

$$r \leq \frac{\pi^C - \pi^{NC}}{\pi^D - \pi^C}$$

r

DISCOUNT RATE

your impatience · ~5–10%

π^C

COOPERATE

\$6B / \$6B · supply & buy continues

π^{NC}

NO COOPERATION

price war wipes profit · **≈ \$0**

π^D

DEFECT ONCE

short-term spike, then relationship collapses

Long-term reward ÷ one-time temptation. With **\$6B/yr** at stake, the inequality holds easily.

WHO INNOVATES, AND WHY

Two effects, two players, two strategies.

REPLACEMENT EFFECT

Apple — the entrant.

Nothing to lose.

Pre-iPhone, Apple had \$0 smartphone revenue.

Big disruptive bet.

The touchscreen iPhone — bet the company on a new category.

EFFICIENCY EFFECT

Samsung — the incumbent.

Now has profits to protect.

Incremental gains: better screens, chips, scale.

R&D climbs incrementally.

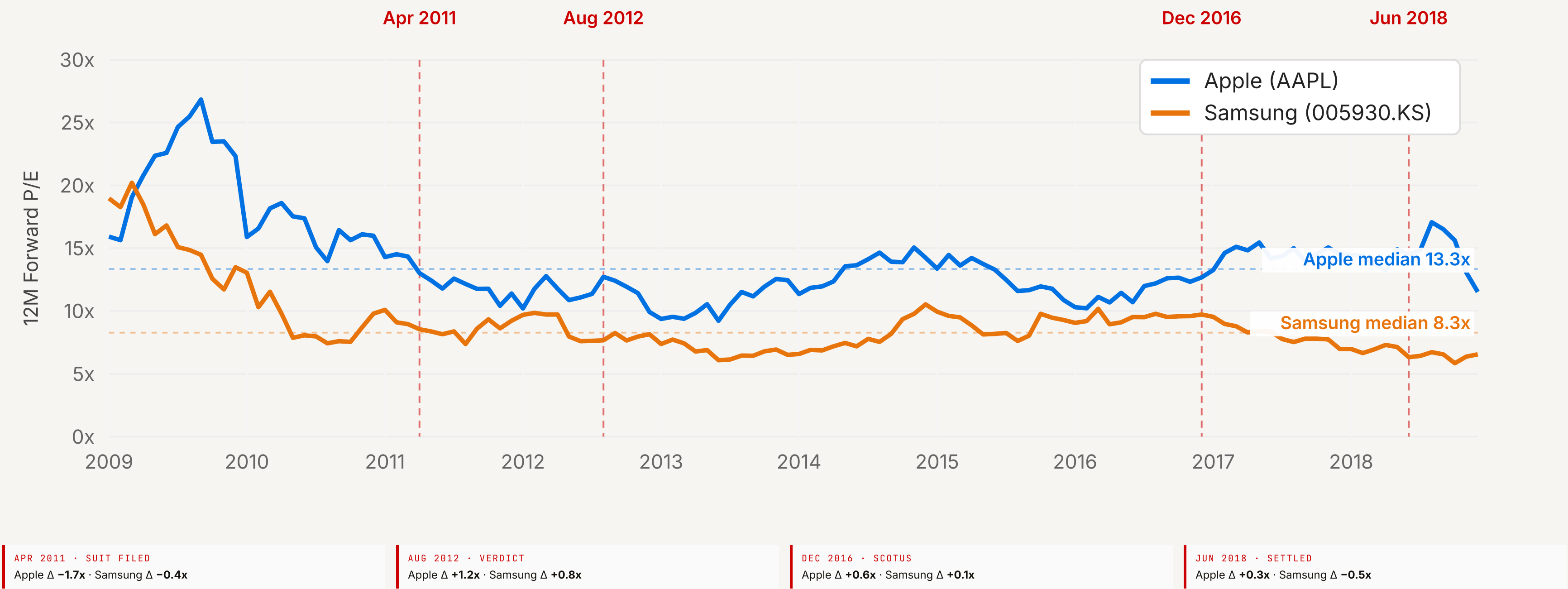
\$8.3B → \$9.1B (2010 → 2011).

THE PRINCIPLE

*"The more you have to lose,
the smaller your bets."*

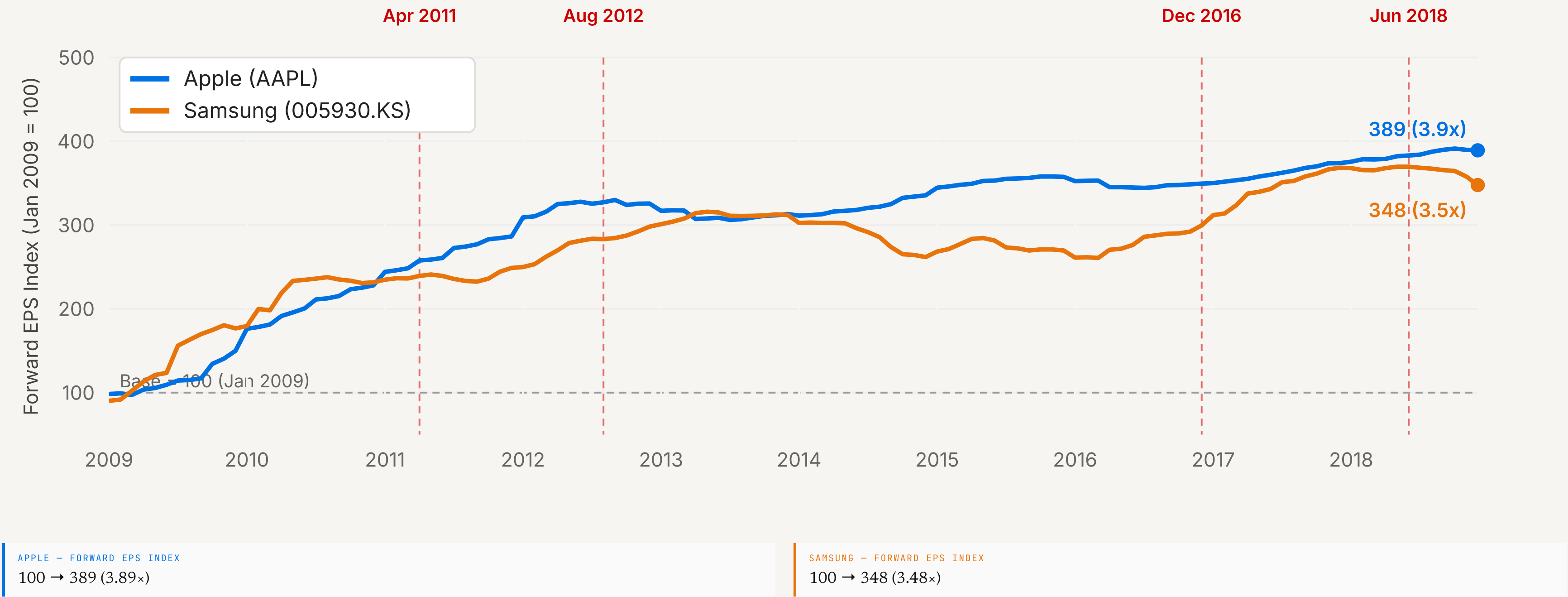
V • WHAT MARKETS SAID

Investors never re-rated either firm on lawsuit news.



WHAT EARNINGS DID

Earnings nearly 4× for both firms — through the entire dispute.



VI • THE VERDICT, FIVE YEARS LATER

Two scorecards. Two stories.

DAMAGES AWARDED
\$539M

8.4% OF SAMSUNG'S 2011 NET INCOME

GOAL OUTCOMES – APPLE VS. SAMSUNG

Apple	
Goal	Outcome
 Damages	\$539M (reduced from \$1.05B) – <i>Partial</i>
 Ban Samsung products	Failed
 Slow Android	Failed (Android hit 52%)

SAMSUNG	
Goal	Outcome
 Become #1 smartphone maker	✓ 90M vs 29M, Q2 2012
 Maintain supply relationship	✓
 Establish Android	✓

CLOSING THOUGHT

“
When companies depend on each other,
patents are bargaining chips, not weapons.
”
Courts move slowly. Markets don’t wait.

EPILOGUE I · SAMSUNG'S SCALE

Six years of growth, in USD billions.

METRIC	2006	2007	2008	2009	2010	2011
Sales	89.4	106.0	110.1	106.8	133.8	148.9
R&D	*	*	*	5.8	7.9	9.0
Net Income	8.3	8.0	5.0	7.6	14.0	12.4
Net Margin	9.3%	7.5%	4.6%	7.2%	10.4%	8.3%
Equity	50.2	60.2	57.1	57.2	77.3	91.9

SOURCE: SAMSUNG AUDIT REPORTS. CONVERTED TO USD AT FRB H.10 PERIOD-AVERAGE KRW/USD: 2006 955.5 · 2007 929.3 · 2008 1,102.1 · 2009 1,276.9 · 2010 1,156.1 · 2011 1,108.3. *R&D BUNDLED IN SG&A PRE-2009 (K-GAAP).

EPILOGUE II • FIVE YEARS LATER

Seven years of litigation, quietly settled.

2016

U.S. SUPREME COURT — 8-0

Design-patent damages should be calculated only on the *copied component* (e.g., the front face), not the whole phone — narrowing the \$399M design-patent portion.

2018

RETRIAL & SETTLEMENT

\$539M awarded. A confidential settlement in June 2018 ended seven years of litigation.

& yet

THE QUIET TRUTH

Despite the war, the supply partnership continued.

